Review Paper

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Shrimp Cultivation in Bangladesh: An Adaptation Strategy or a Maladaptation Strategy

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Abstract

Various studies have been done in Bangladesh to better understand the socioeconomic and ecological effects of shrimp farming in the context of climate change. The available literature covered a wide range of subjects, including farmers' perspectives on growing climate concerns and their effects on shrimp output, shrimp farming as an adaptation technique to salt intrusion, and so on. This paper discusses the problems with shrimp cultivation due to increase in land use for shrimp farming and availability of solutions to move towards sustainable shrimp cultivation. It is seen that Shrimp Cultivation increases inequality among people. Although Rich Farmers and Government are having profit from exporting shrimp, this is just a temporary case. It is necessary to move towards sustainable shrimp farming. ICZM (Integrated Coastal Zone Management) can help Bangladesh in achieving sustainable shrimp farming. Besides, rain-harvesting, new varieties of crops and new varieties of shrimp cultivation can work as sustainable solutions.

Keywords*: Maladaptation, Aquaculture, Shrimp farming, Sustainable, ICZM*

Introduction

Shrimp farming is an aquaculture business that exists in either a marine or freshwater environment, producing shrimp or prawns for human consumption. Globally, shrimp have been farmed for several decades and there currently is production in at least 50 countries around the world, although the industry is concentrated in two major regions, Asia and the Americas. Since the mid-1980s, shrimp, the principal item of frozen food, has been a significant contributor to Bangladesh's national economy. Even though it employs millions of people and generates more than $445 million each year, it has faced several problems. Shrimp farming has been linked to a variety of severe environmental and social consequences, which are preventing the sector's long-term development (Hossain, Uddin & Fakhruddin, 2013). Shrimp farming has a detrimental influence on the cropping pattern and causes deterioration of soil and water quality, depletion of mangrove forest, decrease of local variety of rice and fish, and induces increased salinity leaving no available space for livestock grazing. Nonetheless, shrimp production has been carried out on a huge scale using World Bank and IMF-guided legislative initiatives and incentives with the goal of increasing western exports (Saha, 2017). International organizations have been providing incentives for shrimp farming to improve Bangladesh's economic condition by generating

foreign exchange, but shrimp farming, on the other hand, has resulted in massive pollution of nearby agricultural lands, potentially putting the country's food security at risk.

Over the past two decades, the Shrimp production has expanded in an unregulated way which has caused huge problems to the environment and local people. Thus, our paper will look into this important question:

- Is Shrimp Cultivation Adaptation Strategy or Maladaptation?
- Is it possible to sustainably cultivate shrimp for export in Bangladesh?

As shrimp farming has brought success to the economy and also provided employment to many people in Bangladesh, shrimp cultivation needs to be carried out in a sustainable manner. Therefore, this paper will review the current literature to present the current condition of the Shrimp culture, will explore the accompanying issues and look at potential solutions to make shrimp farming more sustainable.

Methodology

In identifying sources for literature of Shrimp cultivation, multiple databases were used. Initially, Google Scholar was utilized to take a sample of what type of articles were available. Regarding Google Scholar, broad search terms were used to establish a list of research articles that were primary source and peer-reviewed. In the beginning, we used a basic search of Shrimp Cultivation in Bangladesh. From the article titles and research data derived from Google Scholar with that search basis, we were able to use a better list of more refined terms when utilizing other databases like SAGE, Cambridge Core databases. The search terms selected for these literary analyses consisted of: Shrimp Cultivation, Climate Change in Bangladesh, Adaptation Tool, Shrimp Cultivation advantages, Sustainable Shrimp Cultivation and Successful adaptation tool for Bangladesh. These terms were combined in various ways with “AND” commands in the effort to obtain the most narrowly-defined and appropriate articles. Most of these terms were generated due to the initial Google Scholar search, combined with results found from the various database searches. Each of the search terms used were selected due to their appropriateness and relevance in consideration of the purpose of this literature review.

Sources were analyzed according to a number of criteria. First, the source had to be in line with the purpose of the literature review based on the research question of the article. Second, the sources had to be primary source research. Any source that focused on secondary

source research was removed. Third, the sources had to be from a peer-reviewed journal source. In addition to these primary criteria, we also looked for types of journals that would normally include research articles that were theoretically aligned with our purpose. Thus, greater weight was given to articles that focused on Sustainable Shrimp Cultivation. We also examined journals that would be characteristic of their respective fields. Finally, we ensured that the journals used had the most recent publication dates, going back no further than 2000, as Shrimp Cultivation is relatively an old field. After taking these factors into consideration, we analysed the data itself.

When examining the data in the articles, we looked for statements that indicated proper research procedures being conducted. For quantitative data, we looked for larger samples with a degree of diversity in demographic data. In quantitative analysis, we also sought information in the data collection that reflected research questions that was most in line with our purpose, as well as seeking the data with the clearest presentation. Finally, in examining qualitative data, we examined the collection procedures and the demographic data to verify that the researchers had adhered to practices that would ensure reliability and validity.

Literature Review

Advantages of Shrimp Cultivation

Export-oriented shrimp farming, which has grown dramatically in Bangladesh over the last two decades, is one of the country's most important industries. Owing to suitable agro-climatic conditions and good biophysical resources Bangladesh is known to be one of the top places in the world for coastal aquaculture. With over two million individuals working in shrimp growing, selling, processing, and exporting, shrimp culture has provided a diverse range of livelihood prospects for the coastal poor. Over three-quarters of Bangladesh's shrimp farms are in the southwest, with the rest in the southeast. Total annual shrimp output in Bangladesh was estimated to be 75,274 t in 2014–2015, from a total area of 216,468 ha, with an average yearly productivity of 348 kg/ha (Ahmed et al., 2017). Moreover, according to recent studies, Bangladesh's shrimp aquaculture area has grown from around 20,000 hectares in 1980 to approximately 244,000 ha in 2014. The reason behind this rapid expansion has been linked to the enticing international market which has sped up the production of shrimp farming drastically. In addition, international institutions such as the World Bank and the Asian Development Bank, as

well as national legislators, recognized the potential economic benefits of shrimp farming and changed regulations that had aided the fast growth of shrimp culture (Afroz & Alam, 2013). Consequently, one of the primary drivers of large-scale shrimp farming expansion has been the structural adjustment policies which are associated with trade liberalization and trade promotion aimed at promoting exports.(Abdullah et al., 2017).

Critically Looking at the Shrimp Cultivation

Despite the fact that shrimp farming has thrived since its beginning, it has altered the land-use pattern of coastal Bangladesh by replacing conventional agricultural practices. The contribution of shrimp culture to the economy has been emphasized so widely that the long-term sustainability of coastal resources was compromised (Afroz & Alam, 2013). As a result, shrimp farming kept on growing in an unplanned and uncontrolled manner risking valuable coastal resources. The environmental implications of shrimp farming were addressed in various publications in a holistic fashion to portray the current state of coastal land cover change and its resources. Hasan et al (2020) has conducted a study to identify the land use change from 1980 to 2018 for Bagerhat district using remote sensing and GIS technique and discovered that during the last several decades, conversion of agricultural land to shrimp fields led to a mainland cover change in Bangladesh's southern area. Consequently, agricultural land is being reduced in Bagerhat district (which is located in Bangladesh's southwest part) and aquaculture has been growing in this area during the previous 38 years. Saltwater intrusion resulting from the construction of shrimp ponds has been the primary driver of this land-use shift.

Salt Intrusion Destroying Agricultural Lands

Saline water intrusion into fresh agricultural lands deteriorates the soil quality making the lands unsuitable for other crop production. Shrimp production necessitates the use of salty water as an input to the shrimp pond. When land is inundated by salty water for an extended period of time, the water percolates into the surrounding soils, causing changes in soil chemistry. This salinity intrusion has affected the supply of freshwater wetlands which resulted in a rise in gastrointestinal diseases among the population, as well as the loss of diverse crops, poultry, and other livestock. Salt was previously produced in saltpans during the summer by the residents of India’s eastern coast while in winter, they would practice traditional fish and shrimp polyculture or paddy-cum-fish culture. These farming practices never caused any disturbance to the

adjoining agricultural lands. But the current method which requires long-term inundation of the land, prevents free nitrogen fixation, leading to an infertile land in a year or two (Afroz & Alam, 2013). Further, a study has assessed the impacts of shrimp cultivation in Parulia, Satkhira, (Bangladesh) and addressed several changes in the pattern of paddy production due to salt water intrusion. For instance, Farmers previously cultivated local (indigenous) Boro varieties, but now they are cultivating hybrid and high yielding paddy varieties (HYV) such as Jamaibabu 10, BIRI 28, Chini Kanai, Akhter 6, Tejdhan, Hira, and others, as local varieties are unable to withstand excessive soil salinity in this region (Prodhan et al., 2019).

Apart from the changes emerging in the rice growing fields, other crop varieties are at risk of extinction as the production rate is reducing at an alarming rate in Parulia, Satkhira. Several vegetables and fruit trees in the research region, some of which are highly endangered, some are on the verge of extinction. Vegetables like spinach (6.5 percent), celery (5.6 percent), basil (2.8 percent), and arum (2.8 percent) ranked high in terms of extinction, whereas fruits like mango (12 percent) ranked high, followed by blackberry (7.4 percent), jujube (4.6 percent), guava (3.7 percent), and banana (3.7 percent) (3.7 percent) etc (Prodhan et al., 2019). Besides, the conversion of fresh-water fish cultivation into *Gher* (*Gher* refers to the rice fields have been transformed into shrimp aquaculture ponds) has increased the salinity of the water and made it unsuitable for the local fishes such as s Koi (*Anabas testudineus*), Bele (*Periophthalmodon schlosseri)*, Catla (*Catla catla*) etc. As a result, farmers have started cultivating salinity tolerant mono-sex Tilapia, Vetki (Coral), Phaisa, Kurkhullo etc. fishes together with shrimp (Prodhan et al., 2019).

Shrimp Cultivation & Water Quality

Water quality degradation has become a serious concern in the coastal area. In Bangladesh, artificial feeds, agrochemicals, antibiotics, and disinfectants are used extensively in shrimp farming. Excessive usage of these substances causes issues such as toxicity to non-target species, the development of antibiotic resistance, and residue build-up. Owing to the poor management of these shrimp farms, the usage of these chemicals is increasing in an uncontrolled manner. Farmers do not have knowledge about the proper dosage or toxicity levels of the chemicals they use in the shrimp ponds. Further, the pond owners are reluctant to share information about their chemical usage (Afroz & Alam, 2013). This indiscriminate use of chemicals has increased the chance of disease outbreak in the shrimp production, and this can

lead to shrimp mortality and lower yields which has further been confirmed by several studies as outbreaks of bacterial illnesses have harmed shrimp aquaculture in the past few years. However, the chemical composition of the shrimp pond effluents has become a risk for other species as well as humans since the wastes and chemicals from shrimp ponds are directly entering into the environment through water exchange. For a less intensive agriculture, 50-60 million liters of water are necessary to grow one metric ton of shrimp on a more sustainable basis: freshwater accounts for about half of the total (Afroz & Alam, 2013). Hence, the exchange of water has caused problems for surrounding environment and further, caused a scarcity of pure drinking water for the inhabitants (Didar-Ul Islam & Bhuiyan, 2016).

Mangrove Forest Deterioration

Bangladesh's south-west coast is home to the world's biggest compact mangrove forest (Sundarbans Forest). Mangroves play a huge part in the national economy, as well as in the lives and livelihoods of individuals living along the shore (Afroz & Alam, 2013). In Bangladesh, mangrove deforestation is prevalent due to shrimp farming. Because of shrimp farming, the Chakaria Sundarbans in southeast Bangladesh, which contained 8500 acres of mangroves, were deforested. Shrimp cultivation accounts for 1.2 million acres of mangrove destruction. According to a recent study, mangroves in Bangladesh decreased from 452,444 hectares in 1976 to 441,455 hectares in 2010. The mangrove forest is home to a wide range of species, however, is highly crucial for the mitigation of climate change as it stores approximately 56 million tons of carbon, estimated to be valued at least US$1.87 billion global marketplace. Blue carbon ecosystems cover around 51 million hectares worldwide and store 11.5 billion tons of carbon whereas the Sundarbans mangrove forest stores 160–360 t/ha of blue carbon, which consists of 70–158 million tons of blue carbon to Bangladesh depending on plant types and salinity (Chanda et al, 2016). Therefore, the loss of mangrove forest to shrimp culture has resulted in the release of blue carbons. The stock of blue carbon deposited in the soils and biomass of mangrove ecosystems are being exposed to the environment because of the conversion of mangroves to shrimp ponds. As a result, the release of blue carbon can have adverse effects on the coastal community and destroy habitats. Shrimp farming itself will suffer if mangrove forest is no longer present to shelter shrimp farms from tidal surges of storms and other calamities (Ahmed et al., 2017).

Shrimp Cultivation & Socio-economic Changes

From a social standpoint, the expansion of shrimp aquaculture in Bangladesh has exacerbated existing gender and class relations, decreased land security (Hossain et al. 2013), and access to sharecropping opportunities, and increased inequity (Datta 2001; Halim 2001; Belton et al. 2014). The landless have been the most affected by the expansion of shrimp farming, since previously common land has been leased for shrimp aquaculture (Abdullah et al., 2017). Land leasing by powerful shrimp farmers led to a decrease in sharecropping in the coastal area. The small farm owners were forced to lease their lands to shrimp farmers and further, had to sell their property to the shrimp farmers. The government owned lands which were earlier provided to farmers for fishing, vegetation and cattle grazing have been turned into shrimp farms by the influential shrimp farmers. Forcible takeover and counter-occupation of land and shrimp farms, armed attack, shrimp farm poisoning, murder, and severe injuries were reported in the land disputes in the research region (Satkhira, Khulna) (Abdullah et al., 2017). Furthermore, the local people rely on the Sundarbans' resources for their livelihood, but the exploitation of natural resources and the conversion of rice fields to shrimp farms has made life difficult for the poor.

Existing Policies

The management of the overall shrimp production requires specific policy under which the aquaculture can be regulated and controlled in a sustainable manner. Inland fisheries are governed by the Protection and Conservation of Fish Act of 1950, whilst marine fisheries are governed by the Marine Fisheries Ordinance of 1983, which is enforced by the Marine Fisheries Rules. Even though there is no specific policy in place to oversee shrimp farming in the Sundarbans, shrimp farming is subject to current fish and fishery legislation as well as environmental rules in general. However, shrimp farming has grown uncontrollably due to a lack of transparency and convenient adjustments made to shrimp farming for greater revenues.. Therefore, it has become an emergency to adopt a legal framework for sustainable shrimp farming and planning (Afroz & Alam, 2013).

Sustainable Shrimp Cultivation: Integrated Coastal Zone Management

Consequently, to promote sustainable management of coastal zones, effective strategies can be applied to the Sundarbans area to conserve the natural resources of the place. Integrated Coastal Zone Management (ICZM) is a strategy for achieving long-term coastal sustainability. It comprises an interactive planning process resolving the complicated management concerns in the

coastal region. Policy integration, functional integration, and system integration are the three major areas of integration that make up the practical side of ICZM. Policy integration maintains the management program's internal conformity in terms of national and local government policies. The existing policies such as the conservation of fish act or other environmental rules are relevant to shrimp regulation but not exclusively developed for shrimp supervision. Shrimp firming does not cleanly fit within either fishery or agricultural regulations. Therefore, the integration of a separate legal framework for shrimp farming is a successful implementation of the ICZM program. Further, functional integration refers to an institutional framework. In order to address the issue of sustainability, shrimp farming and other users of the coastal zone must work together. It establishes the types and levels of activities that are authorized in compliance with ICZM's aims and objectives. A single agency is required under the ICZM to oversee the zone with the responsibility of shrimp farming. Lastly, system integration will bring all the stakeholders together. In Bangladesh, the most affected communities, the local people and the farmers, are excluded from the creation and execution of governmental policies, such as the location identification of shrimp farming ponds, shrimp farming activities etc. In this regard, a community-based monitoring system might be the most viable strategy to decrease shrimp farming's social, economic, and environmental implications (Afroz & Alam, 2013).

Sustainable Shrimp Cultivation: Rain Harvesting & Mangrove Restoration Projects

However, rain-harvesting can ensure pure drinking water for the inhabitants of the coastal area (Roy, 2017). Furthermore, depending on the monsoon, salt availability, and ideal season, additional concepts have studied the viability of shrimp farming in low, medium, and high saline water. Such zoning for shrimp farming not only maintains natural balance, but also protects the rights of local poor peasants. Besides, programs such as mangrove restoration projects can restore the deforested mangrove forest for the sake of blue carbon storage and mitigate climate change.

Sustainable Shrimp Cultivation: New Variety of Crops

Further, to tackle emerging climatic hazards, the farmers in the southern part have adopted some sustainable farming adaptation measures. To cope with the negative effects of climate change, such as heavy rainfall, seasonal foods, drought, and storm surge, most farmers are adjusting their planting and harvesting times. 90% of farmers are producing high yield

variety (HYV) rice as part of their adaptation strategy to reduce waterlogging and increase soil and water salinity. Further, farmers are encouraged to use saline-tolerant and drought-tolerant crop seeds which can help reduce agricultural production losses. Farmers are now looking for short-duration rice varieties that can withstand floods or high-water levels. (Akter & Ahmed, 2021) Although the use of chemical fertilizers instead of organic fertilizers have been reported for crop production which does not fall under the sustainability shrimp adaptation measures and damage local species.

Sustainable Shrimp Cultivation: New Variety of Shrimp

Lastly, the farming of Pacific white shrimp *Penaeus Vannamei* may bring success to farmers whose farms have become saline and are no longer suitable for growing crops. Due to its rapid development rate, adaptability to high stocking density, reduced dietary protein demand, and tolerance to a broad range of salinity and temperatures, this foreign species has gained enormous momentum in India since its commercial introduction in 2009. The current technology developed by CIFE, Rohtak Centre has shown progress in combating inland saline soils issues in India by the successful cultivation of *Penaeus Vannamei*. Farmers in Haryana and Punjab have been urged to increase production of this new shrimp species in order to bring a blue revolution to the region (Lakra, Reddy & Harikrishna, 2014).

Table 1 below shows the summary of the literature we have used for this study. Most of the authors not only described the disadvantages of shrimp cultivation, but also gave solutions to the problem. The table belows shows all the work of the authors used in this paper in a nutshell.

Table 1: A brief overview of the existing literature on Shrimp Cultivation

Reference	Disadvantage of shrimp farming	Possible solutions for sustainable shrimp cultivation
Ahmed et al., 2017) **(Chanda et al, 2016)**	Mangrove forest destruction and release of blue carbon	
(Afroz & Alam, 2013).	Soil quality deterioration caused by long-term inundation of the land	Integrated Coastal Zone Management (ICZM)
(Didar-Ul Islam & Bhuiyan, 2016) **(Roy, 2017).**	toxicity levels of the chemicals deteriorating the water quality shrimp mortality and lower yields (disease outbreak)	rain-harvesting can ensure pure drinking water (Roy, 2017)
(Prodhan et al., 2019)	Loss of vegetation diverse crops (GIS) and poultry and other livestock	Cultivation of hybrid and high yielding paddy varieties (HYV)
(Hasan et al, 2020)	Loss of fish diversity	Production of salinity tolerant mono-sex fish species (Tilapia, Vetki (Coral), Phaisa, etc
(Abdullah et al., 2017) **(Belton et al. 2014)**	existing class relations as well as gender inequalities have been intensified.	
(Akter & Ahmed, 2021)		- production of saline-tolerant and drought-tolerant crops - short-duration and high yield rice varieties are being used
(Lakra, Reddy & Harikrishna, 2014)		successful cultivation of a new species named *Penaeus Vannamei (White shrimp) has shown promise.*

Discussions

Shrimp cultivation has been used as an adaptation strategy that is used to adapt to climate change. For a poor country like Bangladesh, Shrimp cultivation proved economically beneficial. So, shrimp cultivation started increasing, substituting some other important crop cultivation like

rice, paddy- which are important for the livelihood of local people. From an economics' perspective, when the production of a good starts increasing, there will be a time when the benefit achieved from the good will start increasing at a decreasing rate- Diminishing Marginal Product. When shrimp cultivation substituted rice cultivation during the 1980s, farmers found it helpful as they were financially benefited. Then the cultivation of Shrimp became an important topic due to its playing role in increasing export earnings. Thus, more and more land is being used for shrimp farming.

Shrimp Cultivation as Maladaptation

The rapid increase of shrimp cultivation has prompted both short and long-term terrible environmental impacts, counting biological imbalance, environmental contamination and disease spread. This has furthermore caused a steady decrease in other agricultural and homestead crops, several fruit and woody trees, and homestead vegetables. The unplanned and unregulated shrimp farming ultimately gave rise to numerous socio-economic and ecological impacts. Shrimp farming affects soil and water quality in the cultivated land. The release of saline water from shrimp ponds additionally increases the salinity in neighboring agricultural lands. The unregulated seizure and conversion of traditional farm land to shrimp aquaculture has put food security at risk and left many of the vulnerable people of the coast without alternative livelihoods. Shrimp farming can be considered as "good for the rich and bad for the poor", particularly with regard to food and livelihood security. Shrimp farming is deteriorating Mangrove forest, increasing blue carbon in the environment and as a result disrupting the habitats of coastal people.

Stakeholders in Shrimp Cultivation

Who is the winner in Shrimp Cultivation? - To answer this question, we need to look at the related stakeholders. These stakeholders include: Rich farmers, Poor farmers, Government body & International body.

First, we look at the rich farmers. *Rich farmers* are the high-income households included in shrimp cultivation.. They are the one with the land for shrimp farming and are trying to expand their land by reducing other food-crops and increasing shrimp farming. *Poor farmers* are the lower-income households who had lands for rice, wheat and other food-crop cultivation. As shrimp cultivation increases salinity, these people had to give away their land to the rich farmers.

Higher-income households were able to derive both larger absolute and relative income from the shrimp sector than the lower-income households, mainly because they had more land initially and used that to acquire ownership or access to most remaining land. This has exacerbated existing inequities so they are greater for shrimp aquaculture than for other livelihoods such as fishing or farming. The expansion of shrimp aquaculture has largely happened by converting agricultural land into shrimp farms. Often this agricultural land was acquired from the poorest people, leaving them even more vulnerable because they have been forced into poorly paid labour.

Over recent decades the shrimp aquaculture industry has experienced rapid growth and has taken a key position in the economy of Bangladesh, becoming the second largest export industry after garments. The expansion was driven by the high profit potential of shrimp farming and attracted a wide range of investors, ranging from individual farmers converting paddy fields to multinational companies and corporate business owners investing in large-scale improved extensive and semi-intensive shrimp farming. Due to its potential to improve the economy, the development of the shrimp sector is regarded as an important means of reducing poverty and boosting pro-poor growth, through increased production and exports. Thus, we can say that Government bodies are also benefited by the shrimp cultivation.

Looking into the international bodies, in the beginning, the increase in trade was due to the production of shrimp cultivation being a solution to salt-intrusion. Now that international bodies know the disadvantages of Shrimp Cultivation in Bangladesh, in the near future it can be expected to see a drastic loss in the export of shrimp cultivation. Countries with sustainable shrimp farming will have a better chance in continuing their economic growth by shrimp cultivation, while Bangladesh will be losing the game due to its unsustainable shrimp farming techniques.

Government and Rich farmers are the winners in the short run. However, with the increase in competition, the profit from shrimp industries will be falling rapidly in future. The countries with sustainable shrimp production will be able to continue the increasing economic benefit, while countries like Bangladesh will fall behind. Thus, in the long run, the government and rich farmers are also going to be the losers.

Regarding cost selected per hectare, well-managed shrimp farming is the most profitable sort of land use within the coastal zones of Bangladesh. Overall, it has furnished the country with

an essential possibility for monetary improvement at the macro level. Shrimp cultivating is unsustainable because the negative externalities are precise, endemic, and irreversible. Shrimp generates a financial gain, employment, and overseas trade accordingly the advantages exceed the associated costs. There are negative environmental and social externalities. However, that can be dealt with via policy intervention. This is often a moderate view making an endeavor to limit the considerations. At the national policy level of Bangladesh, the typical perception is that shrimp farming is not any more drawn out as the fundamental reason for ecological degradation. It is the haphazard and impromptu methodology of shrimp farming that is more accountable for environmental and social degradation of the coastal community.

Cost-Benefit Analysis: Success or Failure

Currently, Shrimp Cultivation is still a success. But, the question is: *How long?*

Looking from the stakeholder's perspective,

From the perspective of the International *Body*, now the international community is getting to know about the disadvantages of shrimp cultivation in Bangladesh and trying to move out from investing. In the near future, there will be less and less investors.

For the *Bangladesh government body*, the economy will be harmed a lot when the investors of the shrimp industry starts decreasing. The economic growth due to shrimp cultivation will start decreasing, which is not good for a developing country like Bangladesh.

For *Rich Farmers*, it is a success. They are benefitting from the shrimp cultivation, as they are the land owners. They are the one earning from shrimp cultivation. However, as mentioned before it is just a short-term success. Once the international body moves out their investment, it will not continue to be a success.

For *Poor Farmers*, it is not a success currently and will continue.

Thus, from the current situation Shrimp cultivation is successful in case of Cost-benefit analysis. However, it will not continue to be successful in the future.

Sustainable Shrimp Cultivation

The Bangladesh government has not taken any specific policies to solve the problem of Shrimp Production. However, ICZM implementation can help Bangladesh move into sustainable shrimp farming. Implementation of ICZM can create a separate legal framework for shrimp cultivation. Supervision of shrimp farming under ICZM and ensuring participation of

stakeholders including farmers and local people under the program can help sustainable Shrimp cultivation. A local monitoring system can help to reduce the environmental impact of shrimp cultivation on the non shrimp farming land. The local people at the same time will be supportive of Shrimp farming, as it will affect them less. Policies should be taken to implement ICZM in Bangladesh, to sustainably develop its economy.

Some other sustainable solutions to Shrimp cultivation can include,

Rain Harvesting: Policies regarding rain-harvesting can help poor farmers to save their land and produce their crop with clean water which does not have salt and thus can work against salt-intrusion.

Mangrove Restoration Projects: Mangrove restoration projects should be introduced by the government in order to save mangrove forests and reduce deforestation.

Saline-Tolerant Crop: New varieties of crops like high yield variety (HYV) rice can be produced as an adaptation strategy, which are saline-tolerant and drought-tolerant and as a result will be less harmed by shrimp cultivation.

Farming of New Variety Shrimp: The existing variety of Shrimp in Bangladesh has a number of disadvantages. However, Pacific white shrimp *Penaeus Vannamei* has given positive results in case of quality and is seen as a solution to the Blue revolution in India. Cultivation of such species can also be tried in Bangladesh.

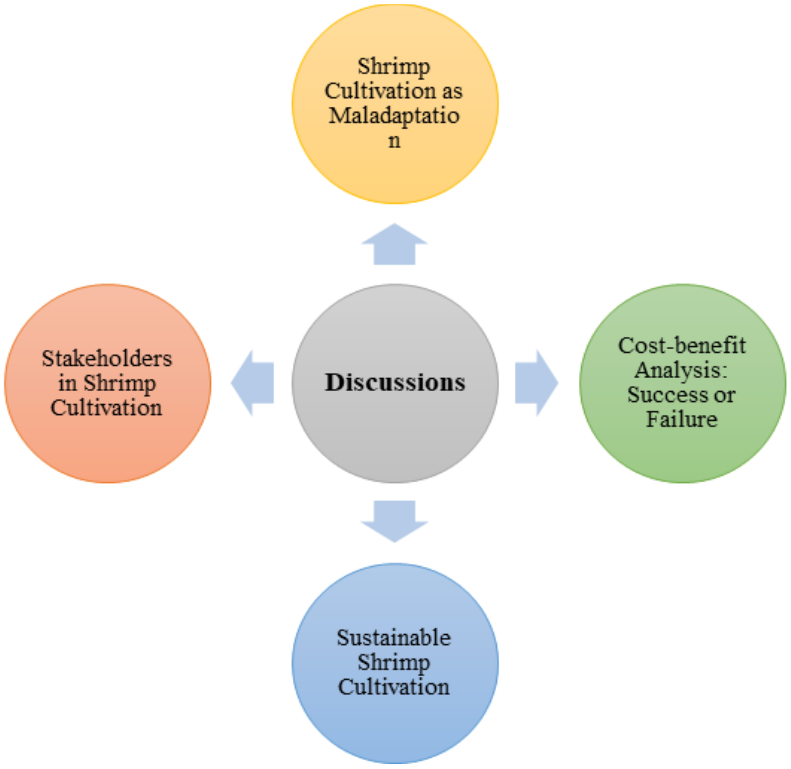


Figure 2 : Discussions in a Nut-shell

Recommendations

Shrimp cultivation is facing high competition in the international market. Moving towards sustainable shrimp farming is very necessary to survive in the market and at the same time continuing the economic growth.

Recommendations for Policymakers

Policymakers can look into the following measures to improve the shrimp industry and make it more sustainable:

- All shrimp farms should be taken under registration for better monitoring that may lead to safe production;
- Inputs of the shrimp farm should be selected very carefully that will not create any harm to the environment;
- To improve the shrimp quality, farm should be practiced standard aquaculture health and better post harvest handling should be practiced;
- Promote diversification into other brackish water aquaculture species like mangrove crab;

- Bangladesh can introduce new shrimp species *Penaeus Vannamei*, which will reduce production cost and increase production like other Asian countries.
- Moreover, the Government should implement ICZM; a step towards sustainable shrimp cultivation.
- Government should introduce Mangrove restoration projects, in order to restore mangrove forest and reduce deforestation.
- Policies to increase production of saline-tolerant and drought-tolerant crop seeds should be taken, which can help reduce agricultural production losses

Recommendations for Future researchers

Although this study has looked into the current disadvantages and solutions of Shrimp Cultivation in Bangladesh. There can be more disadvantages of Shrimp farming in near future. New solutions regarding Sustainable Shrimp Farming can also be introduced in Asian countries. Future researchers are motivated to look into the disadvantages and solutions more carefully in order to help policymakers to come up with policies that are useful and have socio-economic benefits.

Conclusion

Shrimp farming in Bangladesh has expanded rapidly following huge demand in the international market. This fast development is however accompanied by lack of adequate planning and regulation at the national policy level. This unplanned and haphazard shrimp farming is gathering considerable debate due to its negative environmental and socio-economic consequences. Sustainability of shrimp farming is of paramount importance for socio-economic development of southwest coastal Bangladesh. Here shrimp farming is predominantly traditional, which is more vulnerable to climate change. Lack of understanding exists regarding the adaptation measures of local shrimp farmers in response to emerging impacts of changing climatic variables. As there are no specific policies to supervise shrimp cultivation, this important economically beneficial product is not supported by local people and farmers. ICZM in this case, can play a significant role in Bangladesh in moving towards sustainable farming and mitigating climate change.

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